

A Novel CNN-based Autoencoder with Channel Feedback for Intelligent Maritime Communications

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Abstract—Driven by the rapid growth of maritime business, the research of reliable maritime communications has attracted great attention from both academic and industry. This paper proposes a novel convolutional neural networks (CNN) -based autoencoder with channel feedback (CNN-AE-CF) for intelligent maritime communications with complex and changeable environment. A CNN-AE-CF is comprised of CNN layers, which inherits the breakthrough characteristics of CNN, such as generalization, feature learning, classification, and fast training convergence. In the CNN-AE-CF, we introduce a feedback channel and a feedback decoder at the transmitter to decode the feedback signals. Then, the transmitter combines feedback signals with input signals for secondary encoding. We leverage Rician fading channel to simulate the marine environment, in which CNN-AE-CF is trained. Finally, simulation results illustrate the superiority in terms of reliability and robustness. The proposed autoencoder is promising for intelligent maritime communications.

Index Terms—AI, CNN, channel feedback, autoencoder, maritime communications.

I. INTRODUCTION

Driven by the rapid growth of maritime business, like unmanned navigation, resource exploration and disaster relief, high reliability maritime communication systems are required to ensure the safety of ship navigation [1]. The marine environment is complex and the wireless networks are not fully covered, which lead to a lag in maritime communication networks compared with terrestrial communication networks. Therefore, the realization of intelligent maritime communications is very challenging with respect to the following two aspects. One is to be adaptive to the complex and changeable marine environment. Another is the requirement of high reliability for maritime communications.

Deep learning (DL), a branch of machine learning (ML) and artificial intelligence (AI) [2], has been applied in the field of wireless communications, including modulation, encoding, decoding demodulation and channel estimation of communication systems [3]- [6]. The key idea is to represent transmitter, channel, and receiver as a deep neural network that

can be trained as an autoencoder [7]. In order to achieve high-speed and secure information transmission, the autoencoder is required to encode and decode efficiently [8]. T. O'Shea *et.al.* proposed a communication system based on the concept of autoencoder optimized through end-to-end learning in [9]. Nan Wu *et.al.* proposed a CNN-based end-to-end learning framework toward intelligent communication systems in [10], which can work intelligently with arbitrary block length and support different throughput.

Furthermore, channel feedback was applied to the design of maritime autoencoder to encode the transmitted signals twice, which can improve the reliability of maritime communications. B. Dai *et.al.* focused on studying the impact of channel feedback on the transmitter and channel feedback has been shown to help improve the security capability of the transmitter namely reliability in [11]. B. Dai *et.al.* investigated that the security of binary model can be enhanced if there is a noiseless feedback link from the receiver to the transmitter in [12]. Therefore, the channel feedback is expected to optimize the performance of the whole communication systems.

In marine environment, multipath effect is often used to describe the characteristics of marine channel. Marine channel mainly includes direct path and reflection paths [13], which is generally described by dual ray model [14]. Motivated by the desire of deep integration of maritime communications and the technological breakthrough of convolutional neural networks (CNN), B. Lin *et.al.* proposed a novel OFDM autoencoder featuring CNN-based channel estimation for internet of vessels in [15].

In this paper, we propose a novel CNN-based autoencoder with channel feedback (CNN-AE-CF) for intelligent maritime communications, in which Rician fading channel is used to describe the marine multipath effect. Next, the encoder and decoder are designed based on CNN, which is used to extract the features of the signals, to ascend and descend dimensions. Then, we take advantage of feedback channel and

feedback decoder to decode the feedback signals. Moreover, the transmitter combines feedback signals with input signals for secondary encoding. In addition, CNN-AE-CF is trained in Rician fading channel. Simulations illustrate its block error rate (BLER) performance and loss function in marine environment. Finally, the results show that CNN-AE-CF can quickly converge to learn complex signals representation in the training process. Therefore, the proposed autoencoder is provided with strong adaptability and reliability to the marine environment.

II. SYSTEM MODEL

A. Marine Channel Model

To design a maritime communication system with high speed, high reliability, full coverage and low cost, it is vital to model an effective maritime communication channel. The multipath effect is caused by the reflections of the sea surface and the atmosphere when the wireless signals are transmitted, which causes the receiver to receive the signals transmitted from different paths. Due to sparsity, the following two types of paths are extremely important in the marine channel. On the one hand, since obstacles are scarcely presented, the line-of-sight path dominates most of the channel links. On the other hand, reflection paths from the sea surface may exist in some conditions [13]. To model this propagation environment, the two-ray model is utilized in general for wireless channel [14]. The path loss of two-path propagation can be expressed [16] as

$$L_{loss} = -10 \log_{10} \left\{ \left(\frac{\lambda}{4\pi d} \right)^2 \left[2 \sin \left(\frac{2\pi H_1 H_2}{\lambda d} \right) \right]^2 \right\} \quad (1)$$

where λ is the carrier wavelength, d is the distance between the transmitting antenna and the receiving antenna, H_1 and H_2 represent the heights of the transmitting antenna and the receiving antenna, respectively.

End-to-end maritime communications paths mainly include direct path and reflection paths, so we adopt Rician fading channel to describe marine channel. The received signals of Rician fading channel include direct path component and multiple reflection path components. Rician k factors is an important parameter, which reflects the ratio of direct component power and multiple reflection components power. The larger Rician k factor is, the better the channel quality is. Therefore, it is significant to analyze the channel model with Rician k factor and different Rician k factors correspond to different marine environment.

B. CNN-AE-CF for Intelligent Maritime Communications

The autoencoder is a feedforward neural network with unsupervised learning algorithm and generally consists of an encoder network and a decoder network. The encoder and decoder include input layer, hidden layers and output layer, in which the hidden layer has important features of original signals. Communication systems generally consist of transmitter, channel and receiver, in which the transmitter codes the source and the receiver decodes the received signals.

This is very similar to the autoencoder framework. Therefore, communication systems can be regarded as a deep neural network and the end-to-end signal transmission process can be regarded as a signal reconstruction process. The proposed architecture of a CNN-based autoencoder (CNN-AE) for intelligent maritime communications is shown in Fig. 1a, which combines the autoencoder with communication systems, by applying CNN to the hidden layers of the autoencoder. On the basis of CNN-AE, we propose a CNN-AE-CF for intelligent maritime communications with channel feedback.

The proposed architecture of CNN-AE-CF for intelligent maritime communications is shown in Fig. 1b. By using CNN to construct the autoencoder, CNN-AE-CF can extract the spatial information hidden in the signals, where CNN works as a general feature extractor to learn how to capture input signal features. At the transmitting end, the transmitter includes a forward encoder and a feedback decoder. The encoder consists of three One-dimensional convolution layers (Conv1D) and one energy normalization layer. The feedback decoder consists of three Conv1Ds. The transmission channel consists of additive white gaussian noise (AWGN) and forward Rician fading channel. There is only AWGN in feedback channel. At the receiving end, the receiver contains a forward decoder, which consists of three Conv1Ds.

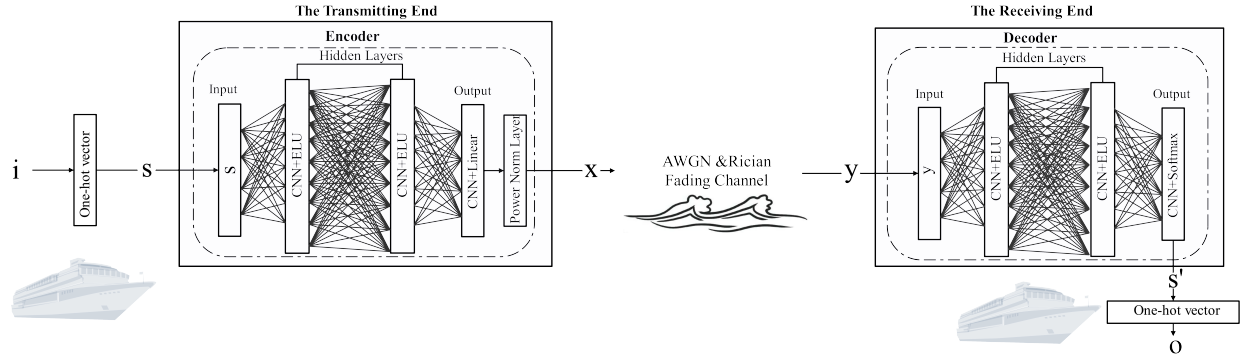
The size of convolution kernel of Conv1D is 1×1 . By adjusting the number of convolution kernels, network data can be ascended and descended dimensions. The number of characteristic channels after convolution is the same as the number of convolution kernels and the values in the convolution kernels are the weight to be trained. With the increase of network depth, the receptive field will be expanded and the extracted features will be more abstract. Conv1D can realize the increase of network depth without increasing the receptive field and introduce more nonlinearity, which will enhance the expression ability of neural networks.

The transmitter sends M messages. The length of transmitting signal s is m , also known as bits per symbol, which equals to $\log_2 M$. Moreover, Conv1D can process the whole symbol sequence s simultaneously, that is, the encoder can process $m \times L$ bits information simultaneously, where L is the block length. In this way, the coding rate of the system is denoted by R , which equals to m/n , where n indicates the channel time slots. The decoded message s' is corresponding to an index with the highest probability p . Moreover, loss function is defined as the mean of measuring performance, namely, the difference between the output and the input. The binary cross-entropy loss function is expressed as

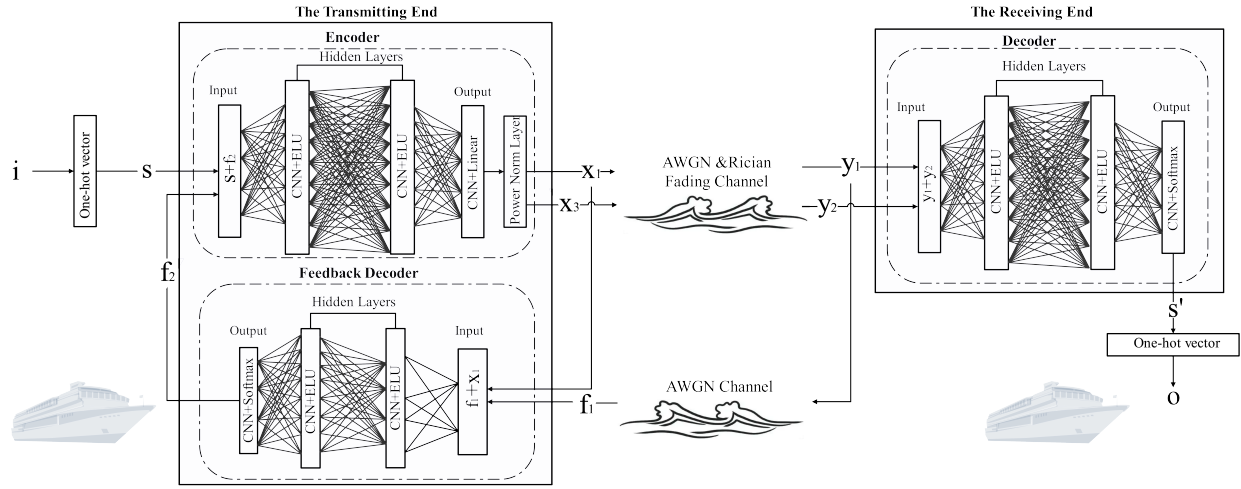
$$F_{loss} = -\frac{1}{N} \sum_{l=1}^N s'_l \log(p(s'_l)) + (1 - s'_l) \log(1 - p(s'_l)) \quad (2)$$

where N indicates training times, s'_l represents the l th element of s' and $p(s'_l)$ is the probability that the output is s'_l . The BLER is expressed as

$$P_e = \frac{1}{M} \sum_l P(s' \neq s|s) \quad (3)$$



(a) The proposed architecture of CNN-AE for intelligent maritime communications.



(b) The proposed architecture of CNN-AE-CF for intelligent maritime communications.

Fig. 1: The proposed architecture of CNN-AE and CNN-AE-CF for intelligent maritime communications.

By leveraging the end-to-end learning mode, the transmitter and receiver can adjust the encoding and decoding parameters to be adaptive to the marine channel.

III. TRAINING OF THE PROPOSED CNN-AE-CF

The structure parameters of CNN-AE-CF are shown in Tab. I. The first column in the table is the layer types. The second column is the activation functions, including exponential linear unit (Elu), Linear, Softmax. The third column is the output dimensions for each layer. L is the block length, n indicates the channel time slots and m stands bits per symbol. The goal of training is to minimize loss function value, which is usually achieved with stochastic gradient descent.

The training process of CNN-AE-CF is divided into three periods, as shown in Algorithm 1. When training the model, it leverages Rician fading channel as marine channel. In the first period, the transmitting end inputs the symbol sequence i . The one-hot vector is chosen to map it to an m bits sequence s . The encoder codes the input signal s by using three Conv1Ds to carry out convolution operation for the whole sequence of

TABLE I: The structure parameters of CNN-AE-CF.

Layer type		Activation function	Output dimensions
One-hot input		None	$L \times 2^m$
Encoder	Conv1D	Elu	$L \times 256$
	Conv1D	Elu	$L \times 256$
	Conv1D	Linear	$L \times 2n$
	Power Norm Layer	None	$L \times 2n$
AWGN & Rician fading channel		None	$L \times 2n$
AWGN channel		None	$L \times 2n$
Concatenate Layer		None	$L \times 4n$
Feedback Decoder	Conv1D	Elu	$L \times 256$
	Conv1D	Elu	$L \times 256$
	Conv1D	Softmax	$L \times 4n$
Concatenate Layer		None	$L \times (4n + 2^m)$
Encoder	Conv1D	Elu	$L \times 256$
	Conv1D	Elu	$L \times 256$
	Conv1D	Linear	$L \times 2^m$
	Power Norm Layer	None	$L \times 2n$
AWGN & Rician fading channel		None	$L \times 2n$
Concatenate Layer		None	$L \times 4n$
Decoder	Conv1D	Elu	$L \times 256$
	Conv1D	Elu	$L \times 256$
	Conv1D	Softmax	$L \times 2^m$
One-hot output		None	$L \times 2^m$

Algorithm 1 CNN-AE-CF Training**Input:** Source one-hot vector s **Output:** Sink one-hot vector s'

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1: while Stop criterion not met do
2:   ——Period 1——
3:    $s \leftarrow$  Input Signal
4:    $x_1 \leftarrow$  Encode Source  $s$ 
5:   Channel  $\rightarrow y_1 \leftarrow$  AWGN and Rician fading channel
6:   Feedback Channel  $\rightarrow f_1 \leftarrow$  AWGN channel
7:   ——Period 2——
8:    $f_2 \leftarrow$  Decode Feedback Signal  $f_1 + x_1$ 
9:    $x_2 \leftarrow$  Encode Feedback Signal  $f_2 + s$ 
10:  Channel  $\rightarrow y_2 \leftarrow$  AWGN and Rician fading channel
11:  ——Period 3——
12:   $s' \leftarrow$  Decode Signal  $y_1 + y_2$ 
13: end while

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the signal. The energy normalization layer is used to constrain the signal, which can be expressed as

$$\|x_1\|^2 \leq n \quad (4)$$

The output is the coded signal x_1 , which can be expressed as

$$x_1 = e(a_1 s + b_1) \quad (5)$$

where $e()$ is activation function, a_1 is the weight matrix connecting input layer and hidden layers and b_1 is the bias vector. The encoded signal x_1 is transmitted through n channel time slots and constrained by Rician k factor, code rate, block length, noise and other conditions. In the process of information transmission, conditional transfer probability p refers to the probability of receiving signal y while sending signal x , which is used to describe AWGN channel, Rician fading channel, and Burst channel. We choose AWGN channel and Rician fading channel as transmission channel. The signal through the transmission channel is expressed as y_1 . Signal y_1 is temporarily stored by the receiver. In the second period, the transmission signal y_1 is feedback to the transmitter through feedback channel to generate feedback signal f_1 and feedback channel contains only AWGN. Feedback signal f_1 and encoded signal x_1 are decoded by the feedback decoder to generate feedback signal f_2 , which uses three Conv1Ds to carry out convolution operations. The feedback signal f_2 can be expressed as

$$f_2 = g[a_2(f_1 + x_1) + b_2] \quad (6)$$

where $g()$ is activation function, a_2 is the weight matrix connecting input layer and hidden layers and b_2 is the bias vector. In the third period, feedback decoding signal f_2 is encoded by the encoder to generate the encoded signal x_2 with input signal s . As the first period, f_2 and s first carry out convolution operation through three Conv1Ds and energy constraint through the energy normalization layer. In addition, the encoded signal x_2 is transmitted through n channel time

TABLE II: The training parameters of CNN-AE-CF.

Parameter Name	Set
Optimizer	Adam
Initial Learning Rate	0.001
Loss function	BCE
Network Performance Indicator	BLER
Batch Size	64
Number of Training	250
Training Set Size	23040
Validation Set Size	2560
Test Set Size	64000

slots and the channel is the sum of AWGN and Rician fading channel. The receiver receives transmission signal y_2 and stores it. Next, the decoder decodes the signal y_1 and y_2 by using three Conv1Ds to carry out convolution operation. The output is the decoded signal s' , which can be expressed as

$$s' = d[a_3(y_1 + y_2) + b_3] \quad (7)$$

where $d()$ is activation function, a_3 is the weight matrix connecting input layer and hidden layers and b_3 is the bias vector. We use one-hot vector to map s' to a symbol sequence $\mathbf{o}$ and output it. This is the end-to-end training process of CNN-AE-CF.

For the training model, our goal is to make output signal s' as close as possible to the input signal s , which can be expressed as $\min \sum |s' - s|$.

IV. SIMULATION RESULTS

Considering the multiple paths caused by the relatively low onboard antennas height and local scatters, we illustrate the generalization ability of CNN-AE-CF in marine environment. We analyze the effect of Rician k factors, channel feedback and feedback channel noises on the BLER performance of CNN-AE-CF in marine environment. Then, we compare loss function of CNN-AE-CF with CNN-AE in marine environment. The training parameters of CNN-AE-CF are shown in Tab. II. The size of the training set is 23040, the size of the verification set is 2560, and the size of the test set is 64000. The model is trained by setting the bits per symbol $m = 4$, channel slot $n = 3$, code rate $R = 4/3$, symbol block length $L = 50$, Rician k factor $k = 1$.

A. The BLER Performance of CNN-AE-CF training in Marine Environment

1) *The impact of Rician k factor:* Fig. 2 demonstrates the BLER performance of CNN-AE-CF in Rician fading channels with different Rician k factors. The model is trained by setting Rician k factor $k = 1, 2, 3, 4$. Simulation results show that the value of BLER decreases gradually with the increase of signal to noise ratio (SNR). Besides, with the increase of Rician k factor, the BLER of CNN-AE-CF in marine environment decreases.

2) *The impact of channel feedback:* Fig. 3 demonstrates the effect of channel feedback on the BLER performance of CNN-AE-CF in marine environment. We compare CNN-AE-CF with CNN-AE. The model is trained by setting the background

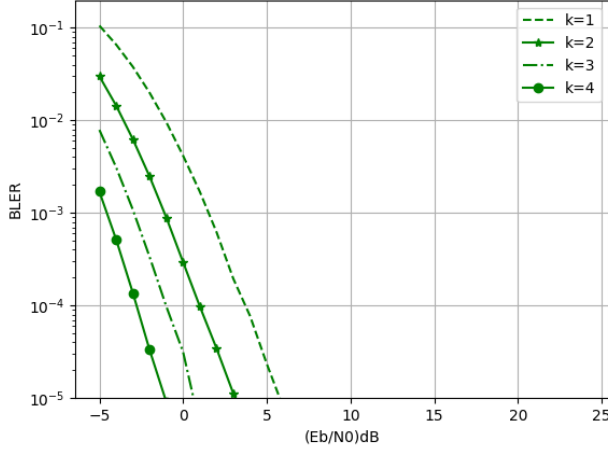


Fig. 2: The BLER performance of CNN-AE-CF in Rician fading channels with different Rician k factors.

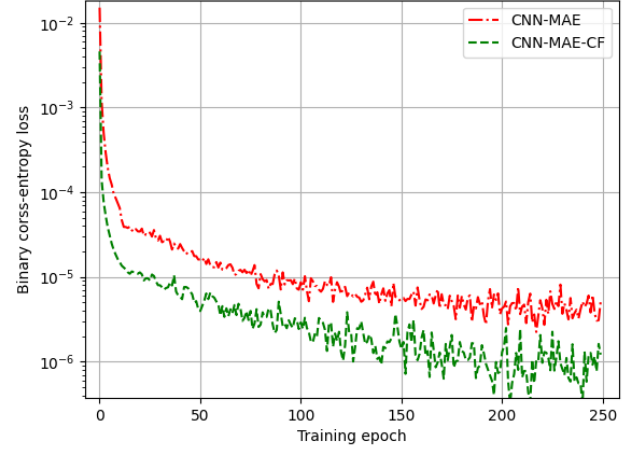


Fig. 5: Binary cross-entropy loss of CNN-AE-CF training compared with CNN-AE in marine environment.

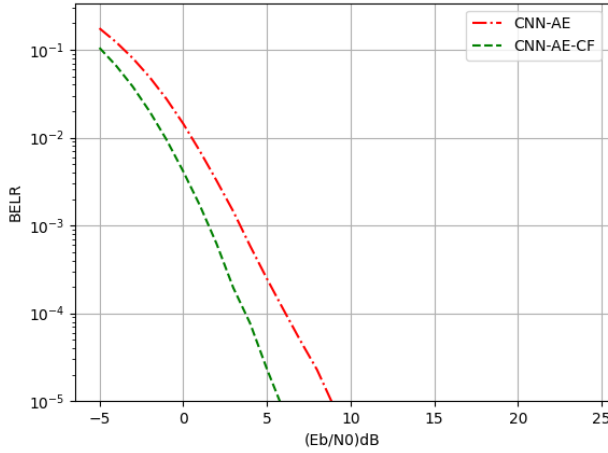


Fig. 3: Effect of channel feedback on the BLER performance of CNN-AE-CF in marine environment.

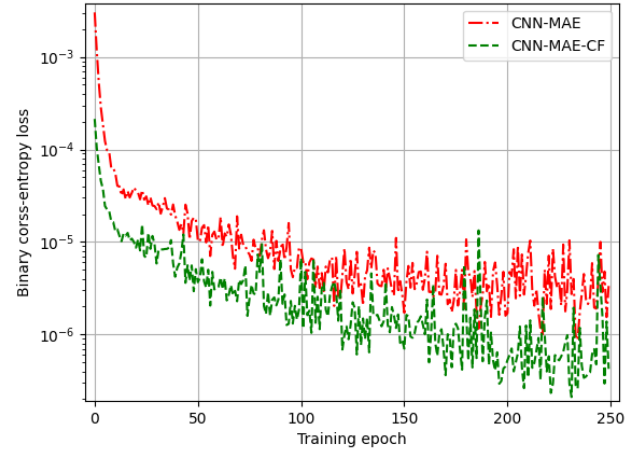


Fig. 6: Binary cross-entropy loss of CNN-AE-CF validation compared with CNN-AE in marine environment.

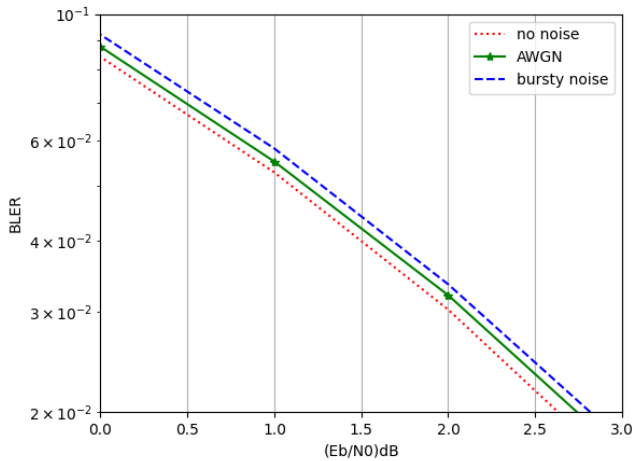


Fig. 4: The BLER performance of CNN-AE-CF under different feedback channel noises in marine environment.

noise in feedback channel as AWGN noise. Simulation result shows that when BLER performance is 10^{-5} , the system SNR improves by about 3.5dB . Simulation results show that the reliability of the system can be improved by adding feedback channel and feedback decoder, in which the feedback signal is decoded for secondary encoding.

3) *The impact of feedback channel noises:* Fig. 4 demonstrates the BLER performance of CNN-AE-CF under different feedback channel noises in marine environment. The model is trained by setting different feedback channel noises. In this experiment, the noises in the feedback channel are selected as no noise, AWGN noise and burst noise respectively. AWGN noise is $W \sim N(0, \rho_1^2)$. The burst noise is $W_{burst} \sim N(0, \rho_2^2)$ with probability ρ , $\rho_2^2 \gg \rho_1^2$, based on the mixed noise model, and $\rho_2 = 10\rho_1$, $\rho = 0.05$ was selected. Simulation results show that the BLER is the lowest when there is no noise in the

feedback channel, and the BLER is the highest when there is burst noise.

B. Binary corss-entropy loss of CNN-AE-CF training in marine environment

We use binary corss-entropy loss to describe the loss of CNN-AE-CF training and validation. Fig. 5 demonstrates the binary corss-entropy loss of CNN-AE-CF training compared with CNN-AE in marine environment. Fig. 6 demonstrates the binary corss-entropy loss of CNN-AE-CF validation compared with CNN-AE in marine environment. The model is trained by setting SNR 12dB. Simulation results show that the binary corss-entropy loss minimum value of CNN-AE-CF is lower than CNN-AE. These illustrate that the reliability has been improved after adding channel feedback. At the same time, binary corss-entropy loss slope of CNN-AE-CF goes down, which indicates that the convergence rate of CNN-AE-CF slows down. The reason is it takes longer time to train the model after adding channel feedback. Therefore, after adding channel feedback, binary corss-entropy loss value of CNN-AE-CF decreases and the convergence speed slows down. Fig. 5 and Fig. 6 show that the model has fully learned the data characteristics, influence of marine channel environment without over-fitting.

To sum up, under different Rician k factors and different feedback channel noises during CNN-AE-CF training compared with CNN-AE training, BLER values are relatively low and the binary corss-entropy loss value is smaller. Thus, CNN-AE-CF has shown higher reliability and more robustness for intelligent maritime communications.

V. CONCLUSION

In this paper, we propose a novel autoencoder, namely CNN-AE-CF, in order to design an intelligent communication system for marine environment. It is suitable for marine environment by employing Rician fading channel as marine channel model to train CNN-AE-CF. It consists of multiple CNN layers to learn intricate signal representations for optimal transmissions, while converging to the optimal solution during training. We leverage feedback channel and add a feedback decoder to the transmitter to decode the feedback signals. Then, the transmitter combines feedback signals with input signals for secondary encoding. With channel feedback, the BLER performance and binary corss-entropy loss of CNN-AE-CF are further reduced, while the number of training convergence iterations does not increase significantly, which is still about 150. In summary, CNN-AE-CF has high reliability and robustness, which is promising for intelligent maritime communications.

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